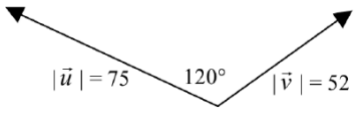
W5 – Cross Product of Vectors Unit 5

MCV4U

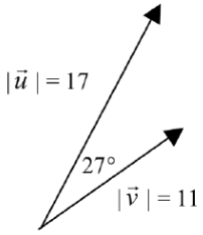
Jensen

1) Determine $\vec{u} \times \vec{v}$.

a)



b)



c) $\vec{u}$= [2, –1, 7], $\vec{v}$ = [2, 1, 3]

d) $\vec{u}$ = [–3, 4, 7], $\vec{v}$ = [4, 3, –5]

e) $\vec{u} = 3\hat{\imath} + 4\hat{\jmath} - \hat{k} \quad \vec{v} = 5\hat{\imath} + \hat{\jmath} - 2\hat{k}$

f) $\vec{u} = 2\hat{\imath} - 3\hat{\jmath} + 7\hat{k} \quad \vec{v} = -\hat{\imath} + \hat{\jmath}$

2) Find a vector perpendicular to each of the following pairs of vectors. Use the dot product to check your answer.

a) [5, 0, 1] and [–2, 5, 8]

b) [1, 4, –2] and [–4, 9, 0]

3) Find a unit vector perpendicular to $\vec{a}$= [6, –2, –3] and $\vec{b}$ = [5, 1, –4].

4) Given $\vec{a}$ = [1, –2, –1], $\vec{b}$ = [2, 2, –1] and $\vec{c}$ = [2, –3, –4], evaluate each of the following:

a) $\vec{a} \times (\vec{b} \times \vec{c}\,)$

b) $(\vec{a} \times \vec{b}\,) \times \vec{c}$

c) $\vec{a} \times \vec{c} - \vec{a} \times \vec{b}$

d) $\vec{b} \times 3\vec{c}$

e) $(\vec{a} \times \vec{c}\,) \cdot \vec{b}$

f) $(\vec{a} \times \vec{b}\,) \cdot \vec{c}$

g) $\left|\vec{a} \times \vec{b}\right|$

h) $\left|\vec{a} \times (\vec{b} - \vec{c}\,)\right|$

5) Use the cross product to determine the angles between the vectors $\vec{a}$ = [2, 1, –3] and $\vec{b}$ = [5, –4, 3]. Consider ambiguous case. Use dot product to confirm or use graphing software to inspect.

6) Determine the area of ΔPQR with vertices of P(3, –2, 7), Q(2, 2, –3), and R(1, 1, 2).

7) Determine the area of the parallelogram ABCD defined by the vertices A(2, -1, -1), B(–4, -2, 3), C(2, 3, 2), and D(8, 4, -2).

ANSWER KEY:
1)a) $-3377.5\hat{n}$ or 3377.5 in to the page **b)** $-84.9\hat{n}$ or 84.9 in to the page **c)** [–10, 8, 4] **d)** [–41, 13, –25] **e)** [–7, 1, –17] **f)** [–7, –7, –1]
2)a) [–5, –42, 25] **b)** [18, 8, 25]
3) $\frac{1}{\sqrt{458}}$ [11, 9, 16]
4)a) [26, 21, –16] **b)** [22, 28, –10] **c)** [1, 3, –5] **d)** [–33, 18, –30] **e)** 13 **f)** –13 **g)** $\sqrt{53}$ **h)** $\sqrt{35}$
5) 96.5°
6) 2.5 $\sqrt{14}$ units^2
7) $\sqrt{1261}$ units^2